

# Intelligent Agent

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# Last time's key concepts

- Agents : An agent is anything that can be viewed as –
  - perceiving its environment through sensors and
  - acting upon that environment through actuators.
- “Rational Agent” as the organizing theme
  - Acts to maximize expected performance measure
- Task Environment : PEAS

# Outline

- Properties of Task environments

# What information do the sensors provide?

- **Fully observable/Partially observable :**
  - A task environment is (effectively) fully observable iff the sensors detect the complete state of the environment
  - "relevant" depends on the performance measure
  - no need to maintain internal state to keep track of the environment
- **Partially observable :**
  - An environment might be partially observable because of noisy and inaccurate sensors or because parts of the state are simply missing from the sensor data .
- **Unobservable:**
  - If the agent has no sensors at all.

# Does the performance depend on other agents?

- **Single agent:**

- . An agent operates independently in an environment, and its performance is unaffected by the presence or actions of other agents.

- **Competitive multi-agent:**

- Improving the performance of one agent directly reduces the performance of another.

- **Cooperative multi-agent:**

- Improving the performance of one agent also improves the performance of the others.

# What effect do the actions have?

- **Deterministic:**

- Outcome of actions (next state of the environment) is completely(fully) determined by the current state.

- **Non-deterministic:**

- Each action has a set of possible outcomes.

- **Stochastic:**

- There is a probability distribution over possible outcomes

# How do the percept change over time?

- **Episodic:**

- Agent receives a **single percept** per episode.
- It's action in this episode doesn't affect future percepts.
- E.g. classification

- **Sequential:**

- Agent receives a **sequence of percepts**.
- The current action affects future percepts.
- E.g. navigation, game playing

# Is the world discrete/continuous?

- Continuous:

- Time, percepts and actions are continuous.
- E.g. driving car

- Discrete:

- Time, percepts and actions are discrete.
- E.g. playing a board game



# How does the environment change over time?

- **Static:**
  - Environment doesn't change unless agent performs an action
- **Dynamic:**
  - Environment changes even when the agent doesn't do anything
- **Semi-dynamic:**
  - Environment is static, but agent's performance score changes over time

# What is the agent's knowledge about the environment?

- **Known:**

- Agent knows **all the rules that** hold in the environment. Can predict outcomes from complete observations.

- **Unknown:**

- Agent **doesn't know the rules** of the environment. Can't predict outcomes

# Summary

- Properties of task environment are
  1. Observable
  2. Agent
  3. Deterministic
  4. Episodic
  5. Static
  6. Discrete
  7. Known

# PEAS for Poker

- **Performance Measure:**

- Winning the game (accumulating the most chips).
- Making profitable decisions (maximizing expected value of bets).

- **Environment:**

- The poker table.
- Other players.
- The deck of cards.



- **Actuators:**

- Actions such as betting, folding, raising, and calling.

- **Sensors:**

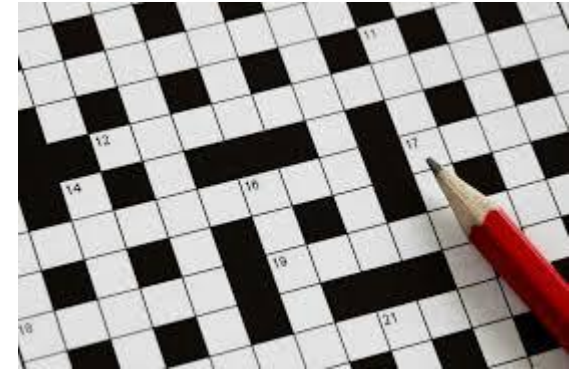
- Visual perception of own cards and community cards.
- Observing opponents' actions and behaviors.

# Example – Poker

- Observable:
  - Agents:
  - Deterministic
  - Episodic:
  - Static
  - Discrete:
- Partially
  - Multi
  - Stochastic
  - Sequential
  - Static
  - Discrete



# Example – CrossWord Puzzle



- Observable:
  - Agents:
  - Deterministic
  - Episodic:
  - Static
  - Discrete:
- Fully
  - Single
  - Deterministic
  - Sequential
  - Static
  - Discrete

# Example – Taxi driving

- Observable:
- Agents:
- Deterministic
- Episodic:
- Static
- Discrete:
- Partially
- Multi
- Stochastic
- Sequential
- Dynamic
- Continuous

# Example – Image Analysis



- Observable:
- Agents:
- Deterministic
- Episodic:
- Static
- Discrete:
- Fully
- Single
- Deterministic
- Episodic
- Semi
- Continuous



# Example – Recipe Execution



- Observable:
- Agents:
- Deterministic
- Episodic:
- Static
- Discrete:
- Fully
- Single
- Deterministic
- Sequential
- Static
- Continuous

# Quiz

In which type of environment does the agent know all the rules and can predict outcomes from complete observations?

- A. Known
- B. Unknown
- C. Partially Observable
- D. Unobservable

# Quiz

Which of the following best describes a **partially observable** environment?

- A. The agent has no sensors at all
- B. The agent can see the entire state of the environment
- C. The agent has noisy or incomplete sensor data
- D. The agent's actions do not affect future percepts

# Quiz

Which property of the task environment is characterized by the environment changing even when the agent doesn't do anything?

- A. Static
- B. Dynamic
- C. Semi-dynamic
- D. Deterministic

# Quiz

Which of the following best describes a **stochastic** environment?

- A. The outcome of actions is fully determined by the current state
- B. Each action has a set of possible outcomes
- C. There is a probability distribution over possible outcomes
- D. The environment does not change unless the agent performs an action

# Exercise

- Develop a PEAS description of the task environment for a face-recognition agent
- Describe the task environment
- Select a suitable agent design for the face recognition agent

# Question ?